

$$\text{Given: } \begin{bmatrix} 4 & -1 \\ -12 & 3 \end{bmatrix} = A$$

$$\text{REF}(A) = \begin{bmatrix} 4 & -1 \\ 0 & 0 \end{bmatrix}$$

Find the Nullspace $N(A)$:

We solve using $Ax = 0$. Using the REF:

$$4x_1 - x_2 = 0 \rightarrow x_2 = 4x_1$$

Let $x_1 = t$ then,

$$x = \begin{bmatrix} t \\ 4t \end{bmatrix} = t \begin{bmatrix} 1 \\ 4 \end{bmatrix}$$

Basis of the nullspace:

So the nullspace is all multiples of $\begin{bmatrix} 1 \\ 4 \end{bmatrix}$.

So, the basis is: $\left\{ \begin{bmatrix} 1 \\ 4 \end{bmatrix} \right\}$